

Exact Differential Equations

A first order differential equation of the form

$$M dx + N dy = 0$$

is called exact if there is a function $f(x, y)$ whose total differential is equal to $M dx + N dy$ that is

$$df = M dx + N dy$$

The general solution of an exact Differential Equations is given by

$$f(x, y) = c$$

Test for Exactness

Let functions $M(x, y)$ and $N(x, y)$ have continuous partial derivatives in a certain domain D .

The differential equation $M dx + N dy = 0$ is exact if and only

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Example 1

Determine whether the following differential equation is exact.

$$y e^{xy} dx + x e^{xy} dy = 0$$

This equation is of the form

$$M(x, y) dx + N(x, y) dy = 0$$

$$M(x, y) = y e^{xy}$$

$$N(x, y) = x e^{xy}$$

$$\frac{\partial M}{\partial y} = e^{xy} + xy e^{xy} = \frac{\partial N}{\partial x}$$

Equation is exact .

Example 2

$$\sin x \cos y \, dx - \sin y \cos x \, dy = 0$$

Example 3

$$\frac{dy}{dx} = \frac{y}{x}$$

Solving exact differential equations

Example 1 $(2xy + 1)dx + (x^2 + 4y)dy = 0$

$$M dx + N dy = 0$$

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}$$

It's necessary to make sure that the DE is exact using the test for exactness

Equation is exact

$\therefore df = M dx + N dy$ වන පරිදි $f(x, y)$ ශ්‍රිතයක් පවතී.

$df = (2xy + 1) dx + (x^2 + 4y) dy$ වන පරිදි $f(x, y)$ ශ්‍රිතයක් පවතී.

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\frac{\partial f}{\partial x} = 2xy + 1 \quad \text{—————} \quad (1)$$

$$\frac{\partial f}{\partial y} = x^2 + 4y \quad \text{—————} \quad (2)$$

(1) න්

$$f = x^2 y + x + \phi(y)$$

We write the system of two differential equations that define the function $f(x, y)$

Integrate the (1) over the variable x

$$\frac{\partial f}{\partial y} = x^2 + 0 + \frac{\partial \phi}{\partial y} = x^2 + 4y$$

$$\frac{\partial \phi}{\partial y} = 4y \Rightarrow \phi = 2y^2$$

Differentiating with respect to y , we substitute the function $f(x, y)$ into (2).

$$f = x^2 y + x + 2y^2$$

but $df = 0 \Rightarrow f = \text{constant.}$

$$\underline{\underline{x^2 y + x + 2y^2 = c}}$$

General solution

Example 2

$$e^{-y} dx - (2y + xe^{-y}) dy = 0$$

Example 3

$$M dx + N dy = 0$$

$$(5x^2 + 12xy - 3y^2) dx + (3x^2 - 2xy) dy = 0 \quad \text{—————} \quad (1)$$

$$\frac{\partial M}{\partial y} = 12x - 6y$$

$$\frac{\partial N}{\partial x} = 6x - 2y$$

This equation is not exact.

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Differential equation is not exact.

Making Exact

(1) $\times x^2$

$$(5x^4 + 12x^3y - 3y^2x^2) dx + (3x^4 - 2x^3y) dy = 0$$

$$\frac{\partial M}{\partial y} = 12x^3 - 6x^2y = \frac{\partial N}{\partial x}$$

A function $\mu(x, y)$ is an integrating factor for a differential equation if $\mu(x, y)[M dx + N dy] = 0$ is exact.

Example

Show that $-\frac{1}{xy}$ is an integrating factor of the differential equation

$$y dx - x dy = 0$$

$$M dx + N dy = 0$$

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial x}$$

Group of Terms

Integrating factor

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$$\mu(x, y)$$

$$y \, dx - x \, dy$$

$$-\frac{1}{x^2}$$

$$\frac{x \, dy - y \, dx}{x^2} = d\left(\frac{y}{x}\right)$$

$$y \, dx - x \, dy$$

$$\frac{1}{y^2}$$

$$\frac{y \, dx - x \, dy}{y^2} = d\left(\frac{x}{y}\right)$$

$$x \, dy - y \, dx$$

$$\frac{1}{x^2 + y^2}$$

$$\frac{x \, dy - y \, dx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$$

Integrating factor to make DE exact

$$(1) \quad \frac{\left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right]}{N} = f(x) \quad \text{a function of } x \text{ alone}$$

Integrating factor $\mu = e^{\int f(x) dx}$

$$(2) \quad \frac{\left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right]}{M} = -g(y) \quad \text{a function of } y \text{ alone}$$

Integrating factor $\mu = e^{\int g(y) dy}$

Multiplying the differential equation by $x^\alpha y^\beta$ yields $x^\alpha y^\beta [M dx + N dy] = 0$. Re-writing in the standard form and using the necessity condition for exactness $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ and comparing the coefficients can find α and β .

Example

$$(x^2 y + y + 1) dx + x(1 + x^2) dy = 0$$

Assignment

Solve the following differential equations

1. $\left(3e^x + \frac{y}{x}\right)dx + (\ln x - 3)dy = 0$

2. $dx - 2xydy = 0$

3. $2xydx + (y^3 - 4x^2)dy = 0$

4. $dx + \left(\frac{x}{y} - \sin y\right)dy = 0$

5. $(8ydx + 8xdy) + x^2y^3(4ydx + 5xdy) = 0$

6. $(4y^2 + 3xy)dx - (3xy + 2x^2)dy = 0$